

**General Principle for
Architecture Design**

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1 SCOPE

This specification outlines the basic principles of the building for the Karabatan Power Plant project of Karabatan Utility Solutions (KUS) located in KARABATAN, Atyrau region, Kazakhstan.

1.1 General

This Specification defines minimum Architectural Design, Construction and other performance requirements for buildings and equipment shelters to be constructed for the Power Plant of the Karabatan Utility Solutions (KUS).

1.2 Design Standards

1.2.1 Regulations

For subjects not covered by regulations, codes or Specifications, the Work shall be based on good workmanship and good economical engineering practice.

The Contractor shall determine by careful scrutiny which of the requirements are more stringent or in contradiction with each other. In all cases, the Contractor shall inform KUS.

This Specification requires compliance with all national and local laws, codes and other authority regulations.

National and local laws, codes and other authority regulations prevail, except where they are less stringent than the requirements given herein, in which case the latter shall prevail.

1.2.2 Project Documents

The latest edition or revision of the following project specifications and drawings relative to building design, specification and construction work at the time of contract award, shall form part of this Specification:

Contractor to develop.

1.2.3 Codes

The Project shall be designed and constructed to meet International standards. Contractor shall execute all required detailed design, construction work and permitting submittals for buildings in compliance with all noted International Institution Standards and Codes as well as in country laws and regulations, and shall obtain required certificates, permits and inspections required by Authorities having jurisdiction for the Project.

The latest editions, revisions and supplements of the following International Institute Codes and Standards relative to building design, materials and component specification and construction work, at the time of contract award shall form part of this Specification :

- International Building Code (IBC) ;

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- International Fire Code (IFC) ;
- International Plumbing Code (IPC) ;
- International Mechanical Code (IMC) ;
- National Fire Protection Association (NFPA) ;
- American Society for Testing Materials (ASTM) ;
- American National Standards Institute Inc. (ANSI) ;
- American Concrete Institute (ACI) ;
- American Institute of Steel Construction (AISC) ;
- American Petroleum Institute (API) .

Additionally, Contractor shall be responsible for ensuring that in-country Standards, Codes and Regulations are adhered to for all Project design and construction work. Where local Kazakh Standards, Codes or Regulations are more onerous than International Standards, Codes or Regulations, they shall apply.

1.2.4 Other Documents

- Blast Effects on Buildings Second Edition ISBN 978-0-7277-3521-8 UFC Structures to Resist the Effects of Accidental Explosions

All Standards and Publications referred to herein shall be those of latest adoption at time of contract award covering work to be performed in accordance with the Specification

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1.3 Definition of Terms

1.3.1 General Definitions

The Following Definitions Shall Be Applied:

"Project" means the ownership, development, design, financing, insuring, construction, operation and maintenance of the Facility and the Client-Developed Infrastructure.

"IPC" means the integrated petrochemical complex to be constructed by the Contractor hereunder at the Karabatan Site.

"Client" means the person named as such in the Agreement and includes its legal successors or permitted assigns

"Contractor" means all and any of the person(s) named as such in the Agreement and includes the legal successors or permitted assigns of such persons.

KUS is the Karabatan Utility Solutions also referred as "The Client".

The Subcontractor is the party which supplies equipment and or services to perform the duties specified by the Contractor.

The Work is the provision of Expertise, Labour, Tools Material and or Equipment as specified in the Purchase Order issued by KUS.

Shall & Should; The word "shall" is to be understood as mandatory and the word "should" as strongly recommended to comply with the requirements of this Specification.

The Engineer is the party appointed by KUS to oversee the Engineering, Design and construction of the Project and to act for and on behalf of KUS.

The word "approved" as used herein shall mean approved by the KUS.

2 DESIGN OF BUILDINGS

2.1 Basis of Design

The contents of this specification shall be adopted by the Contractor and further developed during detailed engineering phases of the Project to meet Client requirements for buildings.

Construction options and materials specifications shall be the subject of technical assessment by Contractor during detailed design activities in order to establish the most suitable structural systems and materials for the buildings to meet the site environmental conditions.

The Contractor shall take due account of the recommendations American Petroleum Institute Recommended Practice API RP 752 when positioning buildings on the plot.

The Contractor may propose for approval alternative arrangements and specifications of components and construction systems to those described herein providing they meet

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all relevant Project standards.

The Contractor shall design and provide all requirements for the building HVAC and fire and safety systems.

All internal finishes and components used in building construction shall be agreed by Contractor with Client during detailed design activities.

2.2 General

- Contractor shall determine in liaison with the Client the type and size of all Project buildings and associated facilities.
- Each building or group of buildings shall be provided with appropriate building services and utilities systems including Heating, Ventilation and Air Conditioning (HVAC), electrical power, telecommunications and water supply, drainage, fire protection and safety facilities etc. to establish and maintain comfortable working conditions for occupants and an efficient operating environment for instrumentation and other technical equipment.
- Contractor shall be responsible for the study of all potential blast loading and building risk assessment information and in liaison with the Client for subsequent definition of hazardous areas used in the blast assessment.
- Contractor shall carry out ergonomic studies and regular building design reviews with Client to ensure that all buildings are fit for purpose and can accommodate required equipment, personnel and functions.
- Following award of Contract, Contractor shall liaise with Client to finalize the preliminary design layouts and elevations of the buildings. Contractor shall allow for minimum two review cycles for preliminary design review, one shall be issued for Client's comments (IFC) and the other shall be issued for design (IFD).
- Client shall be entitled to review at any time the Contractor's detailed designs for buildings to ensure compliance with project requirements. Designs may require to be modified by the Contractor during detailed design activities to the extent necessary to comply with Client requirements.
- Buildings shall be designed and constructed by Contractor in compliance with the component and material specification requirements described below, unless approval of alternative systems is agreed.
- All buildings shall be standardised regarding structural and interior design; and homogenised regarding architectural design.
- The final design and engineering of each building shall be approved by Client prior to the start of building construction.
- The Contractor shall ensure that adequate interior design resource are employed for the project to prepare and agree internal fixtures, fittings, finishes samples and colour schemes with Client.
- Particular care shall be taken in selecting and specifying finishes, textures,

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colours, components, features, and materials.

- Internal finishes specified in this document do not include colours or identified finishes manufacturers -final selection of colours and finishes by Contractor shall be subject to approval.
- Where fire resisting separating walls are required, Contractor shall ensure that the associated supporting structure is protected to the same fire rating, to Client standards and approval. Typically, minimum 1 hour fire rating.
- To mitigate local extremes of temperature the buildings shall be designed to be energy efficient with high levels of thermal insulation.
- In liaison with the Client the Contractor shall provide floor heating systems to Equipment Shelters such as Pumping Stations.
- All building materials and components shall be resistant to pest infestation.
- Each building shall be provided with appropriate fixed furniture, fittings and equipment to reflect function and occupancy type and level. Contractor shall also include for all necessary storage systems in liaison with the Client.
- Where skylights are installed in building roofs (including warehouses) and the roof is designed for access, they shall be guarded (steel grid over skylight) or designed to support the same load as the roof.
- Emergency exits stairways on the process structures handling flammable materials shall be remote from each other and installed at the outside of the structure.
- Building layouts shall be functional with due regards to ease of movement and traffic, and each room and space shall be sized for its intended use. Rooms housing equipment shall include space for future expansion. The amount of space for expansion shall be specified by the Client. Adequate clearance for equipment access now and in the future, shall be provided.
- The climatic design of buildings shall ensure that all electronic equipment can operate efficiently, and in parallel provide an acceptable human working environment.

2.3 Blast Resistance

All buildings shall be designed to meet the appropriate blast resistance criteria given in Project Specification “Design Requirements for Blast Resistant Buildings” 108184-AC-SPC-004.

2.4 Thermal Insulation

All Buildings shall be fully insulated.

Insulation shall be installed in roofs, walls and floors.

The thermal resistance for external walls, roof and floors shall be in accordance with

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national technical regulations for buildings. Or the maximum U value figures given below whichever is the more stringent.

All insulation materials shall be fire-resistant, stable and properly installed to maintain their effectiveness. They shall be rot and vermin resistant; and shall not emit toxic fumes when exposed to fire.

To prevent passage of condensation and to protect the insulation, vapour barriers shall be provided and shall be installed on the warm side of the composite elements. Vapour barriers shall be free of punctures or tears.

The external fabric elements of all buildings shall have a thermal transmittance value (Uvalue) not exceeding the figures given in the table below:

Maximum Building Fabric Uvalues	
Roof	0.20 W/m ² .K
Wall	0.28 W/m ² .K
Ground floor or suspended floor above an unheated space	0.25 W/m ² .K
Windows, roof windows, rooflights, curtain walling and pedestrian doors.	1.8 W/m ² .K
Vehicle access and similar large doors	1.5 W/m ² .K
High-usage entrance doors	3.5 W/m ² .K
Roof ventilators (inc. smoke vents)	3.5 W/m ² .K

2.5 Ventilation of Buildings

Offices and control and administration areas shall have fixed glazing and be ventilated by the air conditioning systems.

Process buildings, workshops and warehouses having a potential for excessive heat in summer months or being at risk of fuel, gas or vaporised chemical leaks from stored materials shall be provided with manually controlled louvered panels and roof vent systems to aid air circulation. The louvered panels shall be selected to prevent wind blown dust entering building interiors.

A smoke indicator opening shall be incorporated at the top of each staircase to permit monitoring from the ground by the emergency services in case of fire.

2.6 Services to Be Supplied to Buildings

The following services are to be provided to all buildings:

- Storm water Drainage ;
- Contaminated Drainage Sanitary services ;

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- Drinking Water Service Water .

Electrical connection and distribution, illumination, telephone, fans, exhaust fans, lightning protection. Air conditioning Fire and gas detection and alarm systems, fire fighting systems.

2.7 Noise Levels

The reverberation time in empty rooms shall not exceed the following values: Office, laboratory, control room 1 second.

Mess room, corridors 1.25 second.

Transmission loss between offices:

Minimum rating of airborne sound insulation between office rooms, assuming closed doors, shall be 40 dB.

Transmission loss between indoor and outdoor:

Minimum rating of airborne sound insulation of external walls against external noise, assuming closed windows, shall be:

Administration buildings 30 dB Control building and laboratory 35 dB Social facilities 40 dB

Minimum ratings as specified in national technical regulations for buildings shall prevail in case they are more stringent.

2.8 Noise Limits

Noise levels in buildings shall not exceed the following limits: Areas in workshops where communication is required. 70 dB(A) Workshop offices and control rooms, not continuously manned. 60 dB(A) Control room, mess room, changing room and toilets. 50 dB(A)

Offices and conference rooms. 45 dB(A).

2.9 Fire Protection

Contractor to develop building fire protection design

3 BUILDING LAYOUT REQUIREMENT

3.1 Minimum Plan Dimensions

- The minimum dimension for the following rooms shall be: Single Offices 12.0 m² per person.

Double (Shared) Offices 16.0 m² total. Management Offices 24.0 m² per person. Canteen Eating Area 2.7 m² per person. Main Corridors 2000 mm wide

Secondary Corridors 1200 mm minimum wide. Toilet cubicles 900 mm x 1200 mm.

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Cleaner room 4.0 m².

- Plant Rooms/Buildings

All electrical, instrumentation and mechanical rooms shall be sized to accommodate the required equipment.

Additional space shall be provided for possible future extension and for ease of circulation of personnel and equipment.

Provide at least two personnel access doors located remote from each other and where possible not in opposite walls.

Equipment doors shall be sized for manoeuvring factory fabrication control boards, cabinets etc. in or out of the building.

Offices, corridor, etc., shall be as for office building.

- Wash and Locker Room

Shower cubicles 2000 mm x 1000 mm

Shower cubicles shall be divided by a 400 mm wide wall into a wet and dry area.

One double locker shall be provided for each shift worker.

Lockers shall be approximately 550 mm wide x 500 mm x 1800 mm high, and mounted on concrete plinth.

Lockers shall include provision for ventilation. 5% Spare lockers shall be provided a hardwood bench for each row of lockers shall also be included. Separate closed rooms for clean and soiled laundry, shall be provided in the wash and locker room. Laundry rooms shall be large enough and be provided with wall mounted shelving.

The domestic premises must be designed according to SNiP RK 3.02.04-2009 («Administration and domestic buildings»).

3.2 Minimum Heights

The following minimum clear heights shall be provided: Offices, toilets, wash and locker room, guardhouse 2600 mm Control room, computer room 3000 mm

Laboratory room and analyzer room 3000 mm Electrical switch room 3000 mm

Fenced area below electrical switch room 1500 mm Corridors and Stairs 2400 mm

All clear heights are from finish floor to soffit of the floor, roof, false ceiling or beams above, whichever is the lower.

3.3 Stairways and Steps

- All stairs both internal and external servicing building shall be constructed of slip resistant materials.
- Localised changes in floor levels of less than 1.0 metres are not permitted within

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buildings without the express permission of the Client.

- The dimensions of steps shall comply with the following rules:
- The maximum rise shall not exceed 180mm ;
- The minimum rise shall not be less than 135mm ;
- The minimum going shall be not less than 280 mm
- The maximum going shall be not greater than 340mm ;

- The minimum width shall be 1000mm.
- The optimum pitch of a staircase shall be when twice the rise plus the going = 600mm.
- Stair flights shall not have greater than 20 steps between landings.
- Landings shall provide a change of direction of at least 90 degrees.
- Landings shall have a minimum width equal to the stairs and a length of not less than 1000mm.
- Handrails and Balustrades shall be provided to all stairs.
- Handrails following the pitch of a staircase shall have a minimum height of 900mm.
- Balustrades shall have a minimum height of 1100mm.

4 DESIGN OF ELEMENTS

4.1 External Walls

- External walls shall be designed to give a neat, clean, aesthetic appearance compatible with the visual aspect of the surrounding.
- Materials selected shall have good thermal insulating properties, be resistant to severe environment and local weather conditions, shall not blemish and shall be easily maintained. Special attention shall be paid to the choice of coatings for the external finishes, particularly metal claddings requiring coatings as a protection against corrosion.
- Walls shall be so designed that they exclude the penetration of rain or snow and prevent passage of condensation into the interior of the building. Particular attention shall be paid at doors, windows and any other openings.
- Vapour barriers shall consist of strong impermeable material, fixed to the warm side of the wall.
- To prevent possible accumulation of gases in the walls, cavity walls or any construction that would encourage accumulation of gas shall not be used in the design of analyzer, electrical and similar buildings. Hollow concrete blocks shall only be used when the hollow cells are completely filled with concrete or mortar.

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- Analyzer, electrical and similar buildings shall be gas-tight. All joints shall be properly executed and sealed including joints at door frames with walls.
- Materials used for closing or sealing any above-ground penetration such as cable or pipe entries shall have the same fire-rating as the particular wall elements.

4.2 Internal Walls

- Materials and construction of internal separation walls shall be chosen for their functions and purposes.
- All internal fire walls shall be built full height, i.e. between structural floor and underside of floor or roof slab above.
- All joints in walls shall be properly executed to exclude voids and crevices to fulfil sound and fire requirements. In fire walls, materials used for closing or sealing penetrations such as ducting, cable trays and pipework shall have the same fire-rating as the walls.
- When a door is installed in the fire-wall, the corresponding fire-rating of the wall shall apply to the door.
- System partitions when installed shall be the removable type and shall extend between the floor finish and the soffit of the ceiling.
- Partitions shall include all doors and windows where required together with conduits and boxes for electrical outlets, switches and data/voice sockets.
- Partitions shall be finished in washable vinyl surfaces.
- Mechanical equipment rooms shall be kept far away from computer and electronic equipment rooms. When this is not possible, a wall, e.g. 300 mm thick, shall be installed to reduce mechanical and electrical interference.
- In wet areas such as toilets, showers, kitchen, etc., moisture resistant finishes shall be used.

4.3 Roofs

- Roofs shall have a minimum slope of 2.5%.
- Any equipment located on the roof shall be supported on concrete plinths. The plinths shall be at least 200 mm above the roof line. When parapet walls are installed, the plinths shall be higher than the overflows in the parapet walls.
- Where frequent traffic is expected for maintenance, walkway in the form of concrete tiles shall be provided. No gravel shall be used to protect the damp-proof membrane.
- Adequate rainwater outlets shall be provided and connected to storm water system.
- Fixed access to all roofs shall be provided either by an external caged ladder or

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through a trap-door in the roof.

4.4 Floors

- All floors exposed to weather, liquid spillage or washing-down operations shall be laid with adequate falls (1:100) to floor drains, gullies or channels, for effective drainage.
- Provisions shall be made for protection of foundation and floor slabs on grade from the effects of a high water table, damp and mineral content.
- All floor slabs exposed directly to ground shall be provided with heavy-duty damp-proof membrane.
- The finished floor level shall be at least 150 mm above the high point of paving.

4.5 Raised Access Floors

- The final design and construction of the raised access floors shall be subject to approval. Cable management void height requirements are notionally 750mm / 1000mm to major equipment room locations.
- The raised floor system shall be a fully accessible, proprietary pre-fabricated system and shall comply with all relevant requirements for loading, electrical conductivity and physical performance. The selected system's structure shall be designed to accommodate specified equipment loads and shall include accessories, cavity barriers and integral floor finishes -all to be of a type recommended by the selected raised floor manufacturer.
- The floor panels shall be formed from heavy-duty composite materials with acoustic properties to prevent impact noise, noise transference and tile rocking. They shall be finished with a factory applied,hard-wearing, antistatic vinyl surface on an aluminium base.
- Floor panels shall be supported by removable stringers and braced, adjustable and lockable steel pedestals, mechanically/chemically fixed to the concrete floor base. Heavy electrical and instrumentation equipment shall be supported by an independent support system.
- Cavity fire barriers, where required shall be of a type compatible with the flooring system. Installation shall be strictly in accordance with the selected floor manufacturer's recommendations.
- The concrete sub-floor and walls below floor level shall be finished with epoxy coatings to prevent dust formation.
- Cable management strategy and required floor or other distribution system / void areas for general administration areas of the Combined Control / Admin / Canteen / Workshop / Stores Building shall be determined during detailed design activities with the Client.

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4.6 Doors

- All external doors shall be single swing, open outwards and shall be fully weather-stripped. All external doors to mechanical plant rooms, telecoms rooms, electrical rooms, battery rooms and instrumentation equipment rooms shall be provided with tight fitting heavy duty pvc seals to prevent gas and dust ingress. All frames shall be fixed into prepared openings within the external walls and shall ensure void-free, solid, waterproof joints caulked with approved paintable mastic. All external doors shall be fitted with door closers appropriate for site wind conditions, door stops and all external emergency exit doors and the active leaves of double doors shall be fitted with door closers, panic latches and door stops.
- Revolving or sliding doors shall not be allowed for emergency exits unless fitted with fused automatic openers that operate in conjunction with the smoke and fire alarms.
- At entrances that lead to areas where dirt is undesirable, mat wells shall be installed.
- All entrances to buildings with an overpressure shall be provided with air locks.
- For access doors, electrically operated doors shall be considered.
- Internal doors of electrical substation, instrument, electrical and mechanical workshops shall be steel doors and frames.
- Door sizing shall allow provision for the movement of the largest equipment module into and out of buildings with the minimum of disturbance to other equipment, fixtures and fittings. Double doors shall be used where required to permit the passage of the largest assembled unit of equipment. Removable transom panels may be utilised to provide the required vertical clearance as required.
- All external personnel and equipment access doors shall be paint finished insulated galvanised steel, complying with all necessary requirements for fire resistance, sound, thermal insulation and security.
- Security rated door assemblies and hardware shall be fabricated with an exterior face of 10mm thick steel plate minimum and an interior face of minimum 1.6mm thick steel sheet or shall be of proprietary steel laminated construction. Frames shall be hot rolled structural steel sections with reinforcing anchors.
- Doors in laboratory areas shall be hygienic type, with radiused pvc / laminate facings and edge details to facilitate good cleaning characteristics.
- Timber doors, used in office areas or in other administration areas, shall be constructed from extruded wood fibre-board with plywood facings, hardwood lippings to all edges and blocking to suit the ironmongery, with a stain and lacquered finished. Vision panels, where required, shall be provided with clear glass panels. All specified door types shall be to Client approval.

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- According to GOST 6629-88 "Internal wooden doors for residential and public buildings", types and construction" the minimum size of a doorway for internal doors shall be 2100x700mm, the width of the door panel is 600mm.

4.7 Fire Doors

- Fire doors shall be self-closing and shall be provided with glazing panel when visibility is required.
- Doors and frames shall be fabricated, using material and construction suitable for the fire rating for which they are intended. For fire rating purposes, door leaves, door frames, hardware and any accessories shall be considered as one complete assembly.
- Fire door assemblies shall comply with the specified requirements herein and shall be supplied complete with relevant test certification and must have compliance certificate.
- Doors shall be smoke tight.
- Fire doors shall be provided with door closers. Closers shall be selected to suite the individual door size/weight and for internal air pressure and external site wind conditions.
- Doors to areas in which 50 or more persons can congregate shall open in the direction of the evacuation.
- Fire doors, shall remain open during normal operation and shall be fitted out with fused automatic closer that operates in conjunction with the smoke and fire alarms.
- Doors in dead end corridors shall be labelled "NO EXIT".
- Fire doors must have compliance certificate.

4.8 Blast-Resistant Doors

"Specification for Design of Blast Resistant Building" (to be developed by Contractor).

4.9 Minimum Door Dimensions

- Single access door 1000 mm x 2100 mm Double door 1800 mm x 2100 mm Equipment door 1800 mm x 2500 mm.
- Toilet and shower doors 700 mm x 2100 mm Doors in removable partition 900 mm x 2100 mm.
- Roll-up door, overhead-section door or sliding door shall be determined during project engineering. Equipment door dimensions shall be increased if required for installation and for bringing equipment in and out of the building.
- Leaf door heights exceeding 2500 mm shall be provided with a removable panel located at the top.

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4.10 Windows

- Limited external windows shall be provided in “Category A” blast resistant buildings which are continuously manned. (Refer to “Design Requirements for Blast Resistant Buildings”).
- Internal windows when provided shall be such that the underside is 1000 mm above the floor finish and the top is in line with the adjacent access doors or 2100 mm above floor finish. Glazing pane shall not exceed 1.0 m² and shall be normal float-glass as described herein.
- All round vision as far as possible shall be provided in the main gate building and guard house. 4.10.4 Window design shall minimise potential heat gain from solar radiation and ultraviolet light transmission by the provision of solar tinted glazing and the provision of external metal louvered sun shades and internal blinds with low heat absorptive properties and stable/ fade resisting colours.
- Non blast proof external windows to the Administration Building and main Guard house shall be anodised aluminium framed, hermetically sealed. All other external windows shall be steel framed.
- Blast resistant windows shall be steel framed galvanised and painted complying with the requirements of Project Specification “Design Requirements for Blast Resistant Buildings” .
- Glazing For Non Blast Resistant Windows
- The type of glass to be used at different locations shall be as follows:

LOCATION	TYPE OF GLASS
Normal internal windows Normal internal doors	Clear tempered safety glass minimum 4 mm thick
Windows between control and operator's laboratory room Internal doors in thick laboratory room	Transparent, laminated glass, 6 mm thick, and one layer tempered glass, 4 mm thick, with a polyvinyl-butylal (PVB) interlayer of at least 2 mm
Fire door	Transparent Georgian polished wired glass, 6 mm thick minimum
Main gate building Guard house	Clear tempered safety glass minimum 4 mm thick

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All external doors and windows	Double glazed transparent with the inner layer consisting of 4mm tempered glass and the outer layer consisting of 10mm laminated glass 4-2-4 with the central layer being a polyvinyl-butylal (PVB) of 2 mm thickness
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4.11 Plumbing

4.11.1 General

The plumbing system shall be designed to include all hot and cold water piping together with all necessary devices, connections to appliances and fixtures, drainage and vent pipe systems including floor drains and cleanouts.

Rainwater outlets from roof and down pipes shall be included and connected to storm-water drainage system when required.

All plumbing work shall be designed in accordance with Eurocodes and local regulations and codes whichever are the more severe.

Toilets for offices, laboratories, canteen, etc.		
Fixture	Male Personnel	Female Personnel
WC	1 for 1 -15 persons	1 for 1 -12 persons
	2 for 16 -40 persons	2 for 13 -25 persons
	3 for 41 -70 persons	3 for 26 -40 persons
	4 for 71 -100 persons	4 for 40 -60 persons
Urinals	nil up to 2 persons	nil
	1 for 3 -10 persons	nil
	2 for 11 -20 persons	nil
	3 for 21 -40 persons	nil
	4 for 41 -100 persons	nil
Hand wash basin with Hot & Cold Taps	1 for 1 -10 persons	1 for 1 -10
	2 for 11 -40 persons	2 for 11 -30 persons
	3 for 41 -60 persons	3 for 31 -45 persons
	4 for 61 -80 etc.	4 for 46 -60 etc.

For processing plants, one WC and one hand washbasin shall be provided for every ten persons on day shift. In addition, one urinal shall be provided for every ten men.

Each toilet area in industrial and utility buildings, including canteens, shall have a tap

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with hose connection and floor drain.

All wet areas shall be provided with floor drains; the number shall be governed by the size of the wet areas.

In the non processing section of the plant (i.e. washrooms in workshops) showers shall be provided at the rate of one per twenty persons. Water troughs shall be provided at a length of 600 mm for every six persons.

In processing plants, for every five operators in day shift one shower shall be installed and 1200 mm length of water trough with two taps. In dirty plants, for every three operators in day shift one shower shall be installed and 800 mm length of water trough with one tap.

For female personnel the same numbers shall be used.

Each washroom shall have a tap with hose connection and a floor drain. The locker room shall be provided with a tap with hose connection and sufficient floor drains.

Each battery room except those with sealed batteries shall be provided with an eye-wash and a floor drain.

Each cupboard shall be provided with a bucket sink, a hose connection, a floor drain, racks for brooms and other cleaning/polishing equipment and shelves for small material.

4.12 Hot and Cold Water System

4.12.1 General

- All water piping shall be arranged to drain to low points and to provide for air elimination at high points.
- Mains, risers and branch connections shall be arranged to permit expansion and contraction without undue stress in the piping system.
- In Offices, Control rooms, Meeting Rooms and the like all lines shall run above the false ceiling and/or chased into walls to minimize the amount of exposed piping. When lines are chased into wall, plastic protection shall be installed to prevent corrosion of lines due to plasterwork. Plastic protection can be in the form of pipe sleeves.
- All exposed water piping shall be insulated.
- All valves shall be easily accessible for maintenance and/or removal.
- Provide unions at connections to each fixture for easy dismantling and at such other points to facilitate installation. Unions shall be of same strength and manufacture as used in the respective piping systems.
- Water supply systems shall be installed in such a way that contamination is prevented at all times.
- Cross connections between systems are not allowed, except when and where approved by the authorities.

Toxic materials are prohibited and shall not be used in water system.

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4.12.2 Materials of Piping Systems

Piping system shall be made of the following materials as scheduled below and as approved by authorities having jurisdiction.

Pipe Materials		
Schedule	Pipe	Fittings
Hot and cold water piping within building	Copper	Capillary fittings
Exposed final connections to fixtures	Chrome-plated brass	Chrome plated brass compression fitting

4.13 Sanitary and Storm Water Sewer Systems

4.13.1 General

- Drainage piping shall be sloped in accordance with accepted hydraulic design principles. Branch connections to main drains shall not be made in a manner which will permit backflow.
- All vent piping shall be arranged to drain any condensate back to waste piping.
- Materials of Underground Drainage Systems
- All materials shall be in accordance with the Project Wastewater Drainage Philosophy.

4.14 Painting

- Painting and protective coatings shall be selected to suit the environment, the climatic exposure and the use of the element to be painted.
- Paint coatings shall have a minimum life expectancy of 5 years from first application.
- Paint coatings shall be non toxic once cured.
- Paint selection shall take into consideration health, safety and environmental factors.
- All proposed coating materials shall be supplied by a single manufacturer and from one manufacturing batch. Different makes may in exceptional circumstances

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be used but only after Client and manufacturer's approval.

- Materials selected for all types of coatings shall, when possible, be water or low-VOC (volatile organic compound) solvent based.
- Surface preparation and method of application shall be as recommended by the paint manufacturer.

4.15 False Ceilings

False ceilings shall be in A2-S1,D0 materials according to “Euroclass” or equivalent. The “Euroclasses” are defined as follows:

CLASSIFICATION
A1, A2, B: Non-combustible and not very combustible products. Over 20 minutes to flashover.
C: Moderately combustible products. Between 10 and 20 minutes to flashover.
D: Moderately combustible products. Between 2 and 10 minutes to flashover.
E: Moderately combustible products.
F : Highly combustible products (or products whose reaction to fire has not been assessed).
Sub-class
S1: Low smoke production.
S2: Medium smoke production.
S3: High smoke production
Flaming droplets sub-classification
D0: No flaming droplets
D1: Flaming droplets that persist for less 10s
D2: Flaming droplets

TESTS METHODS ARE DESCRIBED IN THE FOLLOWING STANDARDS:		
CLASSIFICATION	SUB-CLASS	TEST METHODS
A1	-	EN ISO 1182 and EN ISO 1716
A2	S: Smoke D : Particles	EN ISO 1182 or EN ISO 1716 + EN 13823 EN 13823 (SBI) + EN ISO 11925-2
B		
C		
D		
E	D : Particles	EN ISO 11925-2

For requirements for lowered ceilings see cl. 7.15 SNiP RK 2.02-05-2009 "Fire safety of buildings and structures. Construction standards and regulations of republic of Kazakhstan"

4.16 Finishes

4.16.1 General

In order to facilitate maintenance, a minimum of different floor finishes shall be used.

Ceiling materials including the complete suspension system shall be non combustible.

In wet areas antislip ceramic tiles shall be used. Where sinks are installed in a wall which is not tiled, tiles shall be provided next to and above sink and shall be at least 600 mm round the sink.

Tiles for analyzer room and laboratory shall be acid-resistant tiles.

Locally available materials meeting the standard requirements shall be considered.

4.16.2 Finishes in different buildings

4.16.2.1 Finishes of Floor, Ceilings and Walls

Buildings	Floor Finish								Ceiling Finish						Wall Finish			
	F1	F2	F3	F4	F5	F6	F7	F8	C1	C2	C3	C4	C5	C6	W1	W 2	W3	W 4
Switchgear Building Switch room							X					X			X		X	
Maintenance Building Mechanical workshop Electrical workshop					X X							X X	X X		X X		X X	
Instrument workshop Tool room		X	X		X X				X			X	X		X		X	
Offices Corridor		X X	X X						X X						X X			
Locker room Washroom	X X									X X	X X				X X	X X		
Toilets Cleaner's cupboard	X X									X X	X X				X X	X X		
Mess Room/Kitchen Sitting area Area around counter Kitchen/pantry Stores	X X X	X	X						X X						X		X X X	
Laboratory Building Laboratory rooms Offices Corridors	X X X								X X X							X X		

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Bottle washing	X															X		
Glass/chemical stores	X										X					X		
Toilet	X								X	X						X		
Guard House/Main																		
Gate Building		X	X					X							X			
Guards room		X	X					X							X			
Corridor	X														X			
Toilet																		
Substation (MCC)																		
Switch room						X								X			X	X
Waiting House												X					X	
Waiting room Toilets		X						X				X				X		
Firefighting Deluge					X							X					X	
valve house																		

LEGEND	
F = Floor Finishes	
F1	Vitreous clay tiles, min. 100 x 100 mm, with ditto skirting 100 mm high. For stairs including nosing, wet areas antislip tiles shall be used. Laboratory, analyzer and battery rooms, tiles to be acid resistant.
F2	PVC tiles with PVC skirting. Stair fitted with PVC tiles shall have antislip PVC nosing.
F 3	L inoleum with hardwood skirting.
F4	Wall-to-wall carpets, with hardwood skirting.
F5	Steel-trowel finished green concrete with a carborundum surface finish (2.5 kg/m ²) and a 50 mm high sand cement skirting.

F6	Steel-trowel finished green concrete with a 50 mm high sand cement skirting and a dust-proof epoxy-resin based sealant (two layers) total min. dry film thickness 200-250 µm.
F7	Raised floor with factory-applied antistatic vinyl.
F8	Antislip ceramic tiles.
C = Ceiling Finishes	
C1	Suspended prefabricated acoustic panels of mineral fibre.
C2	Suspended plaster board, plastered and painted.
C3	Plastered and painted
C4	Fair-faced and painted (ceiling will be formed by underside of floor or roof construction)
C5	Special metal diffusing ceiling system to prevent glare on the monitor screens. To be determined.
C6	Fair faced if attractive facing elements are available.

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W = Internal Wall Finishes	
W1	Plastered and painted.
W2	Glazed ceramic tiles up to door frame height, rest of wall W1. If ceiling is low, wall tiles up to ceiling height (i.e. in toilets).
W3	Fair-faced and painted.
W4	Fair-faced, if attractive facing bricks/elements are available.

4.17 Heating, Ventilation and Air-Conditioning

The HVAC installation shall be in accordance with the Project HVAC Performance Specification (to be developed by Contractor).

4.18 Electrical

The electrical aux. installation for power, lighting, etc., in buildings, shall be in accordance with the Project Electrical Specification (to be developed by Contractor).

5 DELIVERABLES

5.1 General

The Contractor shall provide the following minimum deliverables:

Specifications for: Architectural Design / Construction Basis, including but not limited to, masonry, architectural metals, wood and plastics, thermal and moisture protection, doors and windows, architectural finishes, architectural specialities, architectural furnishing, plumbing.

Area Plan drawings as required, landscaping, key and master key system description and schedule, General Arrangement and Architectural drawings (including plan views, sections, elevations, etc. of all buildings).

Other: General building design and design development work to Client requirements and agreement, Project Program, Building Functional requirements, Building Performances (including blast, fire and gas, noise, lighting, plumbing, etc.), Material and Finishing Schedules, building equipment.

6 BUILDING DESCRIPTIONS

6.1 General

The building descriptions below are provisional. The Contractor shall determine and agree all requirements for buildings including floor areas and room type / arrangements within buildings during EPC activities based upon design development

in conjunction with Client, Project staffing levels, Facility equipment location / service requirements and relevant Project Standards and Codes.

All noted construction methods and materials are provisional. The Contractor shall ensure that all construction proposals include materials proven as suitable for site environmental conditions/climate. High levels of thermal insulation will be required throughout to relevant Project Standards and Codes.

Contractor shall agree all proposals for external building fabric including design of external walls/potential use of cavity wall construction, design of roofs for snow loading etc. with Client during detailed design activities.

The Contractor shall determine and gain approval from the Client for the layout, structural approach, material selection and architectural finishes to be applied to every building in his scope of supply.

6.2 Administration Buildings

- This building is part of the Administration Complex.
- The Contractor shall design the building administration accommodation on staffing levels to be agreed with Client during detailed design activities. This building shall include administrative and engineering department areas, individual manager offices, single and double office space, large conference rooms with audio visual and video conferencing facilities, area meeting rooms, covered atrium, emergency response room, archive, ancillary rooms, vending areas, computer facilities, seating areas and all necessary mechanical plant, communications, W.C and support facilities, including passenger / goods lifts. The selection of finishes to be applied shall be of first quality and subject to Client approval The Clients conceptual layout of the Administration complex is detailed on drawings.
- The building shall be security zoned to control access to all facilities contained therein.
- The design of the administration section of the complex shall provide for disabled access and washroom facilities.

6.3 Medical Centre -First Aid

- This building is part of the Administration Complex.
- The buildings shall be designed to contain the Site medical facilities and shall have direct access to a covered emergency vehicle parking area.
- Medical facilities shall include Doctor's room, Nurse and treatment rooms, recovery room with separate W.C, emergency treatment room and appropriate storage and W.C facilities. Stretcher access to these facilities will be required

at all times. A waiting area is provided for personnel with adjacent reception, patient records and pharmacy dispensing.

6.4 Workshops

- These buildings are part of the Administration Complex.
- The building shall be designed to accommodate site Piping, Rotating Equipment, Electrical and Instrumentation workshop facilities with ancillary offices, storage, break room, locker and access to washrooms.
- The structure shall be robust and suitable for the regular movement of heavy loads with the expected occasional accidental impact.
- The workshops shall be equipped with the following overhead cranes. Piping Workshop 2.5 Tonne working load limit.
- Rotating Equipment Workshop 5.0 Tonne working load limit Electrical & Instrumentation Workshop 1.0 Tonne working load limit.

6.5 Capital Spares Warehouse

- This building is part of the Administration Complex.
- The buildings shall be designed to accommodate the Facility's principal storage space for capital spares and spare parts on pallets. The buildings shall include a stores counter, main storage areas and small parts storage areas with adjacent offices, mess rooms and access to WC / locker accommodation.
- The storage racking layout, type and materials movement / automated equipment to warehouse areas shall be agreed with the Client.
- The structure shall be robust and suitable for the regular movement of heavy loads with the expected occasional accidental impact by forklift truck etc.

6.6 Maintenance and Planning Offices

- These facilities are part of the Administration Complex.
- The facilities shall be situated adjacent/within the workshops and consist of an E&I asset management office, project offices and archive rooms for equipment manuals and maintenance diagrams etc.

6.7 Fire Brigade and Ambulance Station

- This building is part of the Administration Complex.
- The building will provide vehicle bays for both fire engines and an ambulance each bay fitted with a fast response roller shutter door with an interlinked audible warning siren on opening.
- Adjacent to the vehicle bays there shall be a storage area for emergency equipment, hoses, fireproof suits etc.
- The ambulance bay shall be equipped with internal double doors providing access to the medical centre.
- Above the vehicle bays there shall be office and mess room facilities.

6.8 Canteen

- This building is part of the Administration Complex.
- The Site dining facilities shall provide seating for people and shall be designed to create a pleasant environment for personnel with the utilisation of high ceilings, shaded window areas and quality internal finishes, furniture and fittings. Number of people to be greased later.
- The kitchen areas shall be designed to be efficient, hygienic and easily maintained with a full range of high quality building finishes, stainless steel heating, serving, dish washing/drying and storage equipment.
- Provided fixtures shall include a stainless steel and glass salad bar and general food servers and associated installations to Client requirements.

6.9 Safety Workshop

- This building is part of the Administration Complex.
- The building shall be incorporated into the workshop/warehouse area of the complex adjacent to the Fire Brigade and Ambulance station.
- The building is intended to be of dual purpose, the primary use being a stores/workshop to provide for the repair and maintenance of safety equipment. The buildings secondary use is to provide an area for safety toolbox meetings. The building shall be equipped accordingly.

6.10 Administration and Workshop Locker Rooms

- This building is part of the Administration Complex.
- The facility shall contain separate male and female changing and washroom facilities with double lockers clean/dirty.

- The access to the facility shall be via a badged access control system.
- The layout of the facility shall be such that it provides a natural separation between the “clean” administration area and the “dirty” process area.

6.11 Personnel and Visitors Car Park

- This area is immediately adjacent to the Administration Complex.
- The Contractor shall provide vehicle sun shade structures. The shelters shall have a roof of insulated profiled metal cladding panels over a painted finished, exposed steel-framed structure that shall be designed for snow and wind loads.
- Personnel shelters shall be provided adjacent to the bus parking area. These may be a dual purpose facility also serving as emergency evacuation assembly point shelters.
- The Car Parking shall be designed for approximately 350 cars.

6.12 Main Guardhouse

- Gatehouse buildings shall accommodate security guards and their associated communications / gate control equipment etc, for control of access to specific areas of the Plant.
- Two weighting bridges shall be designed at each side of the guardhouse.
- The design of Guardhouses shall be integrated into the general layout of site entrance to ensure security and properly control the site access.
- The construction of the main guardhouse shall include full provisions for monitoring the plant security fencing, pedestrian turnstiles, vehicle barriers etc.
- The guardhouse shall be equipped with CCTV capable of monitoring the whole plant, telecommunications systems both internal and external. Visitor’s counter/waiting area, alarm systems and gate controls. A mess room and adjacent WC/washroom.
- The entrance and exit roadways adjacent to the guard house shall be provided with lay-by/waiting areas suitable for a articulated vehicle.

6.13 Evacuation Assemble Point Shelters

- The contractor shall provide a series of single storey insulated buildings at evacuation assembly points. Each building shall be equipped with electric lighting and heating and shall have a minimum footprint size of 5.0mx5.0m.

- The total number of such buildings and their final size shall be determined in consultation with the Client and shall be based on a modular size of 8m² floor area for every 20 people predicted to be at an assembly point.

6.14 Control Building Complex

- The Clients conceptual layout of the Control Building complex is detailed on drawings.
- The complex shall house Control Rooms together with associated support facilities and personnel accommodation as required by Client. The complex is essential for the safe shutdown of the plant in the event of an emergency and shall be designed as a “Category A” blast resistance building.
- The buildings shall be occupied permanently by plant operations, engineering and maintenance personnel and therefore shall be designed to create a calm, pleasant working environment.
- The suite of rooms will typically contain control equipment, workstations, racks, panels and consoles shall include a proprietary raised full access tiled floor system to instrumentation rooms. Support accommodation will include offices, meeting room, WC and locker accommodation, break room and mechanical and electrical plant rooms. The complex shall be enclosed with fire resistant construction to in accordance with this specification.
- The building will also operate as an emergency crisis centre thus it shall be capable of operating without reliance on external facilities.
- The Contractor shall provide sufficient resources and employ the necessary relevant specialists needed to provide the detailed design proposals and gain approval. As a minimum the specialists shall include for, ergonomics studies, materials selections, access floor design, lighting design, colour scheme, ceiling design and furniture provision in order to create the most pleasant and appropriate control room environment, in accordance with current industry guidelines and relevant codes of practice.
- All work in relation to control area access floors and lighting shall be fully coordinated and agreed with the Control and Instrumentation Engineers prior to finalisation and installation. The coordination shall including for the agreement desk and work positions.
- The control room and MCC ventilation system (HVAC) shall be designed with a hydrocarbon detector in the makeup air to shut off the flow if the vapour concentration exceeds 5% of the lower flammable limit. The air inlet for the ventilation shall be located in a hydrocarbon free area (minimum of 5m above the building). An easily accessible switch shall be provided in the control room to shut off the HVAC system. The air intake openings shall be provided with explosion dampers (blast valves) designed for the predicted overpressure.

- The MCC, instrument rack room and battery room shall be separated from the main control room with two hour fire resistant walls.
- At the UPS room provide continuous mechanical ventilation over the battery charging area if the batteries are not sealed, sufficient to remove any flammable gasses. Ventilation shall be designed to extract gas from the top of the room.
- Safety shower and eye wash station if batteries are not sealed.
- Where wet cells batteries are used, the UPS controller shall be in separate room from the batteries.
- A mosaic panel that indicates the layout of the combustible gas detectors, oxygen analyzers, fire detection and automatic extinguishing systems, wind speed and wind direction shall be installed in the control room. The alarm status of each device shall be indicated. Alternatively, a dedicated large LCD monitor can be installed depicting the items indicated above. The display shall be redundant on the DCS monitors.
- External facilities shall be limited to mechanical and electrical support plant in separate compounds designed to complement the building design.

6.15 Laboratory + Engineering Technology and Quality Office

- This building is part of the Control Building Complex.
- The buildings shall house the plant sample and analysis laboratory facilities along with ancillary rooms, laboratory offices, maintenance personnel offices, storage, W.C and locker room accommodation. Contractor shall include specialist professional design input to agree accommodation and specification of the Laboratory areas with Client during detailed design activities.

6.16 First Aid Building

- This building is part of the Control Building Complex.
- The facilities will be equipped as minimum with fully equipped cabinet for a physician, equipped rooms to receive injured persons, breathing apparatus equipment, little surgery equipment; basic cardiac simulator; a safe to keep at site a minimum of medicines etc.

6.17 Permitting Area

- This is part of the Control Building Complex.
- The area will contain a permitting office, a permitting counter and a waiting

area.

- The area will be adjacent to the central control room and the internal counter area shall be directly accessible from the control room.

6.18 Control Room Complex Locker Rooms

- This building is part of the Control Room Complex.
- The facility shall contain separate male and female changing and washroom facilities with double lockers clean/dirty.
- The access to the facility shall be via a badged access control system.
- The layout of the facility shall be such that it provides a natural separation between the “clean” administration area and the “dirty” process area.

6.19 Substations

- Substation consists of a switch room on a raised floor and a transformer pit supported on concrete piled foundation.
- The space between the raised floor and grade shall be used as cable lay-down area.
- Fencing shall be provided between the raised floor and grade all around the building.
- Electrical equipment (switchgears) shall be supported on the raised floor. The raised floor, foundation and transformer pit shall be cast in situ reinforced concrete.
- External walls and roof shall be self-supporting precast concrete elements.
- Substation shall have galvanized steel platforms and stairs.
- The transformer pit shall have reinforced concrete walls and bottom slab filled with 30/60 gravel.
- Transformer pits shall be drained via a normally closed control valve to the contaminated sewer system.
- A reinforced concrete or masonry fire wall shall be provided between adjacent transformers.
- Fencing of approximately 2.0 m high from the top of the pit shall be provided all around the transformer pit.
- The fence shall include a 900 mm wide gate with lock for each transformer.

6.20 Drivers Waiting House

This building is part of guardhouse.

6.21 Fire Fighting Deluge Valve House

Deluge valve house shall have load bearing concrete walls, a cast insitu reinforced concrete foundation slab and precast concrete roof elements. Deluge valve house shall be fire rated for two hours.

6.22 Operator Shelters

Where required, these buildings shall be provided to accommodate facilities for plant operations personnel within more remote parts of the process area and shall typically include a maintenance team room, office accommodation, lockers, break room, meeting room, WC / shower and store.

6.23 Satellite Rack Rooms (SRR'S)

Satellite Rack Room buildings shall house an instrumentation equipment room, communications equipment, battery room, UPS room and mechanical plant room.

6.24 Analyser Houses

These buildings are assumed to be packaged units supplied by 3rd party vendors. (Contractor to verify).

6.25 Compressor and Other Equipments House

Pumps, compressors and other equipment shall be provided with high level weather protection and general maintenance access / equipment support. All such requirements shall be confirmed and agreed with Client and equipment suppliers during detailed design activities to suit selected equipment.

6.26 Process Buildings

- The Contractor shall include for insulated building enclosures for process equipment to allow continuous winter operations.
- The structures shall be robust and suitable for the regular movement of heavy loads with the expected occasional accidental impact.

- The design shall incorporate provisions for temporary supports for maintenance access and shall consider access requirements for the replacement of equipment and pipework.

6.27 Product Warehouses

- These buildings shall be designed to accommodate the Facility's storage space for product / bagging lines and process associated materials. The buildings shall include main storage, loading, dispatch areas with offices, mess room WC / locker accommodation.
- The structure shall be robust and suitable for the regular movement of heavy loads with the expected occasional accidental impact by forklift truck etc.

6.28 Chemical Storage

- The building shall be provided as dedicated chemical stores and as secure stores for hazardous materials used on site.
- The internal floors shall have dedicated drains connected to the contaminated sewer system via buffer basin with a manually operated valve normally closed.
- The walls and roof of the building shall have a minimum 2 hour fire resistance and be provided with blow out pressure relief panels.
- Special attention shall be given to the provision of smoke, gas and heat sensors connected to the plant alarm system.

6.29 Logistics Loading and Distribution Buildings

These buildings shall accommodate small scale offices for computer weigh scale terminals and ancillary accommodation to include personnel office, driver waiting area, small kitchen and WC / locker facilities.

6.30 Car Park Shelters and Truck And Drivers Facilities

These Buildings accommodate vehicle driver's amenity facilities within the Site Lorry Park / Container Yard.

Each building shall be provided with a rest room / kitchenette, storage / janitor's room and WC / shower facilities.

6.31 Communications Buildings

These buildings shall contain accommodation for communications equipment and associated battery room.

6.32 Railway Platforms

Railways and railway station/platforms for loading and transporting of products are outside of EPCM Contractor's scope of work. However, the EPCM contractor shall take into account both railway terminals to dispatch the products as described in the Logistic Basis of Design (Logistics BOD).

6.33 Railway Security Building

This building shall accommodate security personnel and associated communications, security lighting, surveillance and gate control of all access to railway areas. The building shall be located in an area adjacent to the line of security fences within a paved area, controlling vehicular access and egress and all required security gates and barriers.

6.34 Waste Water Treatment and Solid and Liquid Waste

The building shall be designed to house the plant water treatment processes and equipment, the water treatment chemicals, the water testing laboratory, offices, locker rooms, WC and mechanical , electrical and telecommunication rooms.

6.35 Canine Compound / Dog Facility Building

This building is deleted.

6.36 Power Plant

The contractor shall provide detailed drawings of the power plant buildings and facilities. The security and fireproofing issues shall be agreed with Client during detailed design activities.

7 SOLID AND LIQUID WASTES

7.1 General

The EPCM contractor shall develop the Client's strategy for waste treatment which is based on the following principles listed by hierarchy: prevention, minimisation, re-use, recycling, energy recovery and disposal.

7.2 Liquid Wastes

Waste waters and domestic sewage will be treated at the treatment plant.

The treated waters will then be released to the evaporation ponds.

Those ponds are out of EPCM Contractor's scope of work.

7.3 Solid Wastes

- The solid wastes may be treated and disposed through a proprietary solid waste treatment system.
- Solid waste includes both medical and domestic waste.
- Licensor's guidelines shall be respected by the Contractor.
- The buildings shall include naturally ventilation and air conditioned storage areas.
- The buildings shall be constructed using full height separating walls in reinforced masonry construction and external walls and roof clad in noncombustible, foam / mineral fibre insulated, profiled metal composite cladding panels with a corrosion resistant finish and integrated Louvre system.
- Electrical and communication rooms shall be provided to Client requirements.
- Floor finishes shall be specified to resist all products in use and will include consideration of appropriate non-slip epoxy based chemically-resistant coatings to ensure robustness of floor. Design will include bunded areas to detailed design with coved floor finishes.
- The ground floor and surrounding access slabs, hard-standing and ramps shall be designed for vehicle usage as agreed with the Client.
- Loading facilities shall be provided and shall be suitable for lorry and fork lift truck loading.
- Insulated steel roller shutter doors with PVC curtains shall be provided as external vehicle access to the building, sized in order to allow articulated vehicles to load/unload as required.
- Emergency eye wash and shower units shall be provided to relevant codes and standards.